

# Advanced Statistical Learning

## Linear Methods for Classification

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Oct 13th, 2025

# Outline

Linear Regression of an Indicator Matrix

Linear Discriminant Analysis

Logistic Regression

Naive Bayes

Separating Hyperplanes

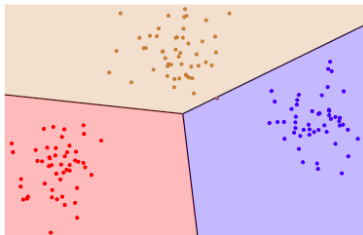
# Some Intro

Want to learn a predictor  $G: \mathbb{R}^p \rightarrow \mathbb{G}(1, 2, \dots, K)$

$G$  divides input space into regions labelled according to their classification.

The boundaries between these regions are termed the decision boundaries.

When these decision boundaries are linear we term the classification method as linear.



# Linear Decision Boundaries

Model the discriminant functions  $\delta_k(x)$  or posterior probabilities  $Pr(G = k \mid X = x)$

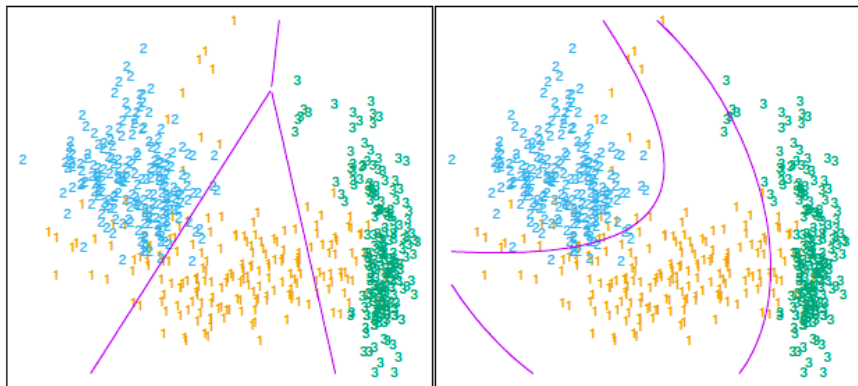
- ▶ If some monotone transformation of  $\delta_k(x)$  or  $Pr(G = k \mid X = x)$  is linear in  $x$ , then decision boundaries are linear in  $x$
- ▶ Linear discriminant analysis
- ▶ Logistic regression

Explicitly model the boundaries between the classes as linear.

- ▶ Separating hyperplane - a normal vector and a cut-point
- ▶ Rosenblatt perceptron model
- ▶ Optimal separating hyperplane - minimize some measure of overlap

# Linear Decision Boundaries - Generalization

Expand variable set  $X_1, \dots, X_p$  by including their squares and cross-products  $X_1^2, X_2^2, \dots, X_1X_2, \dots$



Linear decision boundaries  $\rightarrow$  Quadratic decision boundaries.

# Linear Regression of Indicator Matrix

# Linear Regression of an Indicator Matrix

Training data  $\{(x_i, g_i)\}_{i=1}^n$ , where each  $x_i \in \mathbb{R}^p$  and  $g_i \in \{1, \dots, K\}$

For each  $k$  construct a linear discriminant  $\delta_k(x)$  via:

1. For  $i = 1, \dots, n$ , set

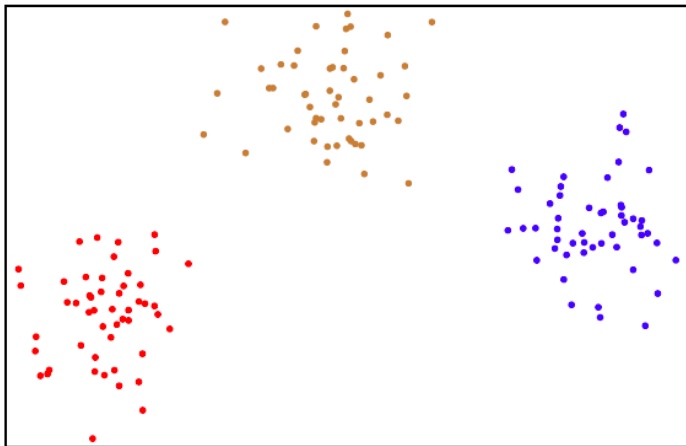
$$y_{ik} = \begin{cases} 0, & \text{if } g_i \neq k \\ 1, & \text{if } g_i = k \end{cases}$$

2. Compute  $(\hat{\beta}_{0k}, \hat{\beta}_k) = \operatorname{argmin}_{\beta_{0k}, \beta_k} \sum_{i=1}^n (y_{ik} - \beta_{0k} - \beta_k^\top x_i)^2$
3. Define  $\delta_k(x) = \hat{\beta}_{0k} + \hat{\beta}_k^\top x$

Classify a new point  $x$  with

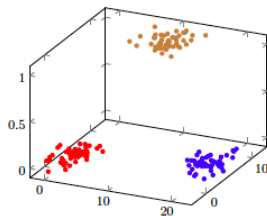
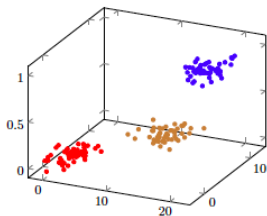
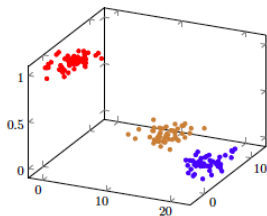
►  $G(x) = \operatorname{argmax}_k \delta_k(x)$

## 3-Class Example

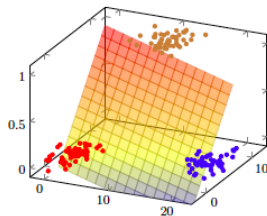
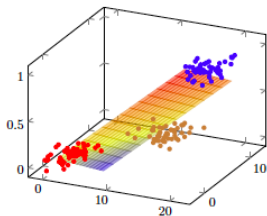
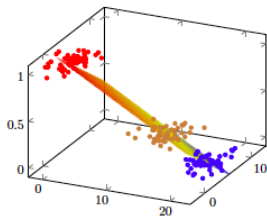


Use linear regression of an indicator matrix to find the discriminant functions for the above 3-classes.

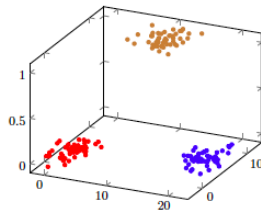
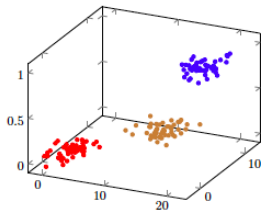
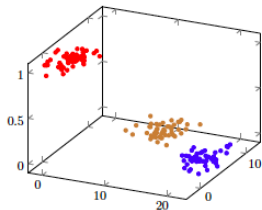
For each  $k$  construct the response vectors from the class labels



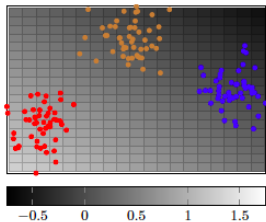
For each  $k$  fit a hyperplane that minimizes the RSS



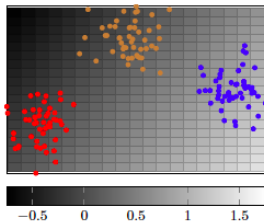
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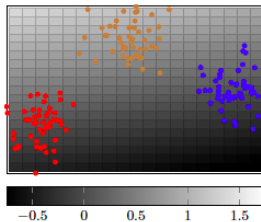
The  $k$  discriminant fns defined by the least square hyperplanes



$$\delta_1(x)$$

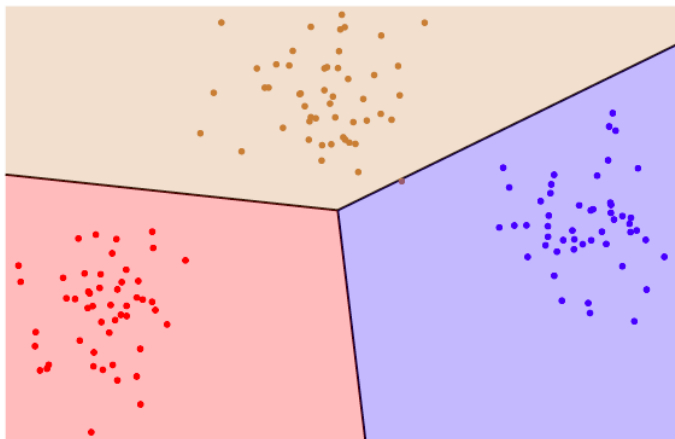


$$\delta_2(x)$$



$$\delta_3(x)$$

# 3-Class Example: Decision Boundary



# Linear Regression of an Indicator Matrix

Rationale?

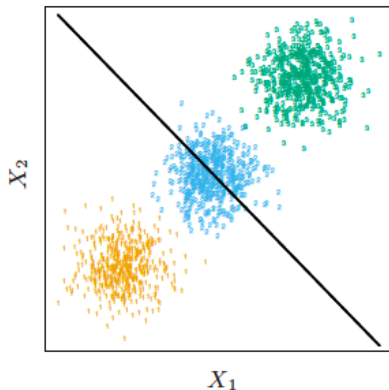
View the regression as an estimate of conditional expectation

- ▶  $E(Y_k \mid X = x) = Pr(G = k \mid X = x)$

Any problem?

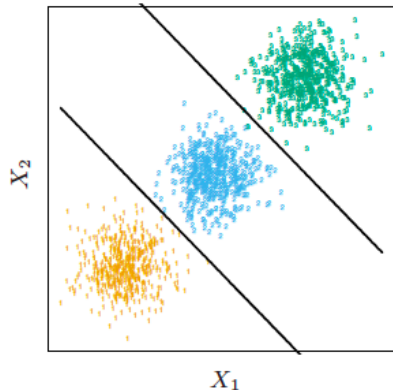
- ▶ How good is the approximation?
- ▶ Predicted values out of  $[0, 1]$
- ▶ Possible remedies?

# When This Approach Fail



Linear regression of indicator matrix:

- ▶ Middle class completely masked

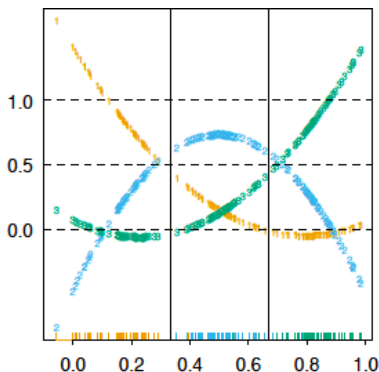
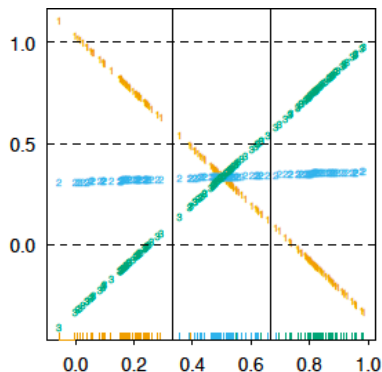


Linear discriminant analysis

- ▶ Well separable boundary

# When This Approach Fail

Linear and quadratic polynomials regression for indicator matrix



To solve "masking" problem when  $K \geq 3$ , a loose but general rule:

- Polynomial term + cross products  $\sim O(p^{k-1})$

# Linear Discriminant Analysis

# Optimal Classification

To perform optimal classification, need to know  $Pr(G \mid X)$ , let

- ▶  $f_k(x)$ : class-conditional density of  $X$  in class  $k$
- ▶  $\pi_k$ : prior probability of class  $k$

By Bayes Theorem

$$Pr(G = k \mid X = x) = \frac{f_k(x)\pi_k}{\sum_{l=1}^K f_l(x)\pi_l}$$

Therefore for classification, having  $f_k(x)$  is almost equivalent to having  $Pr(G = k \mid X = x)$

# Model Class Densities

Methods that are based on specific models of  $f_k(x)$

- ▶ Linear & quadratic discriminant functions use Gaussian distribution
- ▶ Mixture of Gaussian distributions produce non-linear decision boundaries
- ▶ Non-parametric density estimates allow the most flexibility
- ▶ Naive Bayes assumes  $f_k(X) = \prod_{j=1}^p f_{kj}(X_j)$

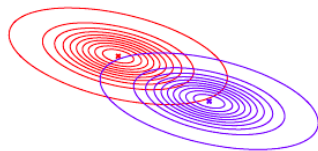
# Multivariate Gaussian & LDA

Model each  $f_k(x)$  as a multivariate Gaussian

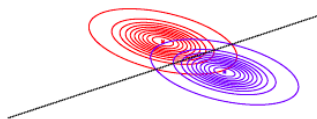
$$f_k(x) = \frac{1}{(2\pi)^{p/2} \sqrt{|\Sigma_k|}} \exp \left\{ \frac{1}{2} (x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) \right\}$$

Linear Discriminant Analysis (LDA) arises in the special case when

$$\Sigma_k = \Sigma \text{ for all } k$$



**class distributions**



**decision boundary**

## In comparing class $k$ and $l$

$$\log \frac{\Pr(G=k|X=x)}{\Pr(G=l|X=x)} = \log \frac{f_k(x)}{f_l(x)} + \log \frac{\pi_k}{\pi_l} =$$
$$\log \frac{\pi_k}{\pi_l} - \frac{1}{2}(\mu_k + \mu_l)^\top \Sigma^{-1}(\mu_k - \mu_l) + x^\top \Sigma^{-1}(\mu_k - \mu_l)$$

The equal covariance matrices allow  $x^\top \Sigma_k^{-1} x$  and  $x^\top \Sigma_l^{-1} x$  cancel out

From the log odds, we see that the linear discriminant functions

$$\delta_k(x) = x^\top \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^\top \Sigma^{-1} \mu_k + \log \pi_k$$

are an equivalent description of the decision rule with

$$G(x) = \operatorname{argmax}_k \delta_k(x)$$

# LDA - Practicalities

In practice we do not know the parameters of the Gaussian distributions, and will need to estimate them using our training data

Let  $n_k$  be the number of class  $k$  observations then

- ▶  $\hat{\pi}_k = \frac{n_k}{n}$
- ▶  $\hat{\mu}_k = \sum_{g_i=k} \frac{x_i}{n_k}$
- ▶  $\hat{\Sigma} = \sum_{k=1}^K \sum_{g_i=k} \frac{(x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^T}{n - K}$

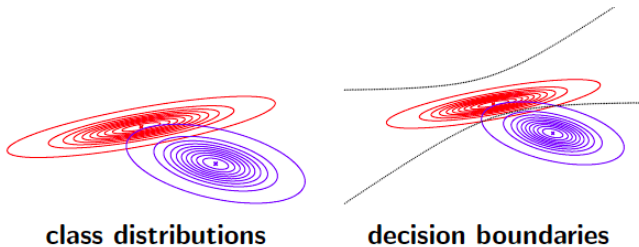
Note: for  $k = 2$  there is a simple correspondence between LDA and linear regression of indicator matrix, which is not true for  $k > 2$ .

## When $\Sigma_k$ Not Equal

If the  $\Sigma_k$  are not assumed to be equal, then the quadratic terms remain and we get quadratic discriminant functions QDA

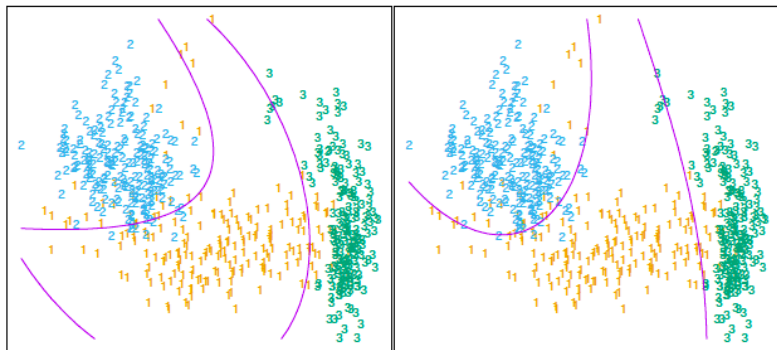
$$\delta_k(x) = -0.5 \log |\Sigma_k| - 0.5(x - \mu_k)^\top \Sigma_k^{-1}(x - \mu_k) + \log \pi_k$$

In this case, the decision boundary between classes are described by a quadratic equation  $\{x : \delta_k(x) = \delta_l(x)\}$



# Quadratic Discriminant Analysis

Left: quadratic decision boundaries found using LDA in the five dimensional space  $X_1, X_2, X_1^2, X_2^2, X_1X_2$ ,

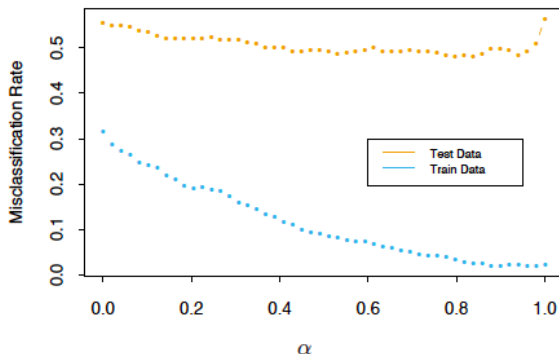


Right: quadratic decision boundaries found by QDA

# Regularized Discriminant Analysis

$$\hat{\Sigma}_k(\alpha) = \alpha \hat{\Sigma}_k + (1 - \alpha) \hat{\Sigma}, \alpha \in [0, 1]$$

- ▶  $\hat{\Sigma}_k(\alpha)$  Regularized covariance matrices
- ▶  $\hat{\Sigma}$ : pooled covariance matrix



Can also replace  $\hat{\Sigma}$  by  $\hat{\Sigma}(\gamma) = \gamma \hat{\Sigma} + (1 - \gamma) \hat{\sigma}^2 I$ ,  $\gamma \in [0, 1]$

# Computations for LDA & QDA

Diagonalizing  $\hat{\Sigma}$  or  $\hat{\Sigma}_k$  by eigen decomposition  $\hat{\Sigma}_k = U_k D_k U_k^\top$ , where  $U_k$  is  $p \times p$  orthonormal, and  $D_k$  is a diagonal matrix of positive eigenvalues  $d_{kl}$ . The ingredients for  $\delta_k(x)$  are

- ▶  $(x - \hat{\mu}_k)^\top \hat{\Sigma}_k^{-1} (x - \hat{\mu}_k) = [(x - \hat{\mu}_k)^\top U_k] D_k^{-1} [U_k^\top (x - \hat{\mu}_k)]$
- ▶  $\log |\hat{\Sigma}_k| = \sum_l \log d_{kl}$

Therefore, LDA can be implemented by:

- ▶ Sphere the data with respect to the common covariance estimate  $\hat{\Sigma}$ :  $X^* \leftarrow D^{-\frac{1}{2}} U^\top X$ , where  $\hat{\Sigma} = U D U^\top$ . The common covariance estimate of  $X^*$  will now be the identity
- ▶ Classify to the closet class centroid in the transformed space, modulo the effect of the class prior probabilities  $\pi_k$

# Reduced Rank LDA

# Reduced Rank LDA

$K$  centroids in  $p$ -dimensional input space lie in an affine subspace of dimension  $\leq K-1$

if  $p \gg K$ , there is a big drop in dimension

To locate the closest centroid can ignore the directions orthogonal to this subspace.

Therefore, can project  $X^*$  onto this centroid-spanning subspace  $H_{K-1}$  and make comparisons there

LDA thus performs dimensionality reduction and one need only consider the data in a subspace of dimension at most  $K-1$ .

# What About a Subspace of Dim $L < K-1$

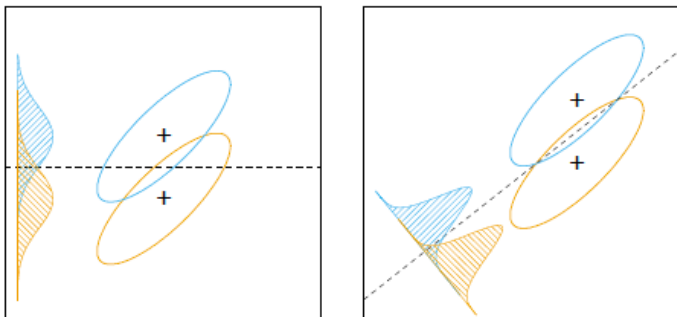
- ▶ Which subspace of dim  $L < K-1$  should we project onto for optimal LDA?

Finding the sequences of optimal subspaces for LDA involves:

- ▶ Compute the  $K \times p$  matrix of class centroids  $M$  and the common covariance matrix  $W$  (for within-class covariance)
- ▶ Compute  $M^* = MW^{-\frac{1}{2}}$
- ▶ Compute  $B^*$ , the covariance matrix of  $M^*$  (B for between-class covariance), and its eigen-decomposition  $B^* = V^* D_B V^{*\top}$
- ▶ The columns  $v_l^*$  of  $V^*$  in sequence from first to last define the coordinates of the optimal subspaces.
- ▶  $l_{th}$  discriminant variable  $Z_l = v_l^{\top} X$ ,  $v_l = W^{-\frac{1}{2}} v_l^*$

# LDA via Fisher Criterion

Find the linear combination  $Z = a^T X$  such that the between-class variance is maximized relative to the within-class variance



# LDA via Fisher Criterion

$W$ : pooled covariance about the means

$B$  : covariance of the class means

Then for the projected data  $Z$

- ▶ The between-class variance of  $Z$  is  $a^T B a$
- ▶ The within-class variance of  $Z$  is  $a^T W a$

Fisher's criterion

$$\max_a \frac{a^T B a}{a^T W a}$$

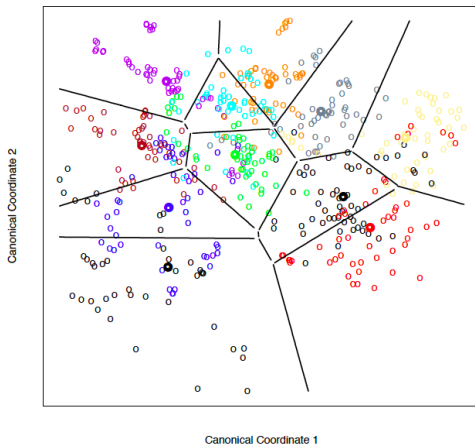
or equivalently

- ▶  $\max_a a^T B a$  subject to  $a^T W a = 1$

$a$  is given by the largest eigenvalue of  $W^{-1}B$ , which is identical to  $v = W^{-1/2} v_1^*$  defined before.

# Classification in Reduced Subspace

Original data have 11 classes and 10 variables



Decision boundaries in two dimensional subspace spanned by the first two canonical variates.

# Logistic Regression

# Logistic Regression

Directly model the posterior probabilities of the  $K$  classes

$$\log \frac{Pr(G=1|X=x)}{Pr(G=K|X=x)} = \beta_{10} + \beta_1^T x$$

$$\log \frac{Pr(G=2|X=x)}{Pr(G=K|X=x)} = \beta_{20} + \beta_2^T x$$

...

$$\log \frac{Pr(G=K-1|X=x)}{Pr(G=K|X=x)} = \beta_{(K-1)0} + \beta_{K-1}^T x$$

Class  $K$  as reference level, all probabilities sum to 1

Each probability lie in  $[0, 1]$

$$\blacktriangleright p_k(x; \theta) = Pr(G = k \mid X = x) = \frac{\exp(\beta_{k0} + \beta_k^T x)}{1 + \sum_{l=1}^{K-1} \exp(\beta_{l0} + \beta_l^T x)}$$

# Logistic Regression

For  $k = 1, \dots, K - 1$

$$\blacktriangleright Pr(G = k \mid X = x) = \frac{\exp(\beta_{k0} + \beta_k^T x)}{1 + \sum_{l=1}^{K-1} \exp(\beta_{l0} + \beta_l^T x)}$$

For  $k = K$

$$\blacktriangleright Pr(G = k \mid X = x) = \frac{1}{1 + \sum_{l=1}^{K-1} \exp(\beta_{l0} + \beta_l^T x)}$$

Linear decision boundary between classes

$$x : Pr(G = k \mid X = x) = Pr(G = l \mid X = x)$$

which is the same as

$$\{x : (\beta_{k0} - \beta_{l0}) + (\beta_k - \beta_l)^T x = 0\}$$

for  $k = 1, \dots, K - 1$  and  $l = 1, \dots, K - 1$

# Logistic Regression Model Estimation

Two-class case, log likelihood can be written as

- ▶  $l(\beta) = \sum_{i=1}^N \{y_i \log p(x_i; \beta) + (1 - y_i) \log(1 - p(x_i; \beta))\}$
- ▶  $\text{logit } p(x_i; \beta) = \log \frac{p(x_i; \beta)}{1 - p(x_i; \beta)} = \beta^\top x_i, \beta = (\beta_0, \beta_1)$

To maximize the log likelihood, obtain the first and second derivative

- ▶  $\frac{\partial l(\beta)}{\partial \beta} = \sum_{i=1}^N x_i (y_i - p(x_i; \beta))$
- ▶  $\frac{\partial^2 l(\beta)}{\partial \beta \partial \beta^\top} = - \sum_{i=1}^N x_i x_i^\top p(x_i; \beta) (1 - p(x_i; \beta))$

Iteratively update the initial value by Newton–Raphson algorithm

- ▶  $\beta^{new} = \beta^{old} - \left\{ \frac{\partial^2 l(\beta)}{\partial \beta \partial \beta^\top} \right\}^{-1} \frac{\partial l(\beta)}{\partial \beta}$
- ▶ Derivatives evaluated at  $\beta^{old}$

# Logistic Regression Model Estimation

In matrix notation, let  $W$  be  $N \times N$  weight matrix with  $i_{th}$  diagonal element  $p(x_i; \beta^{old})(1 - p(x_i; \beta^{old}))$

Then the first and second derivatives can be expressed as

$$\blacktriangleright \frac{\partial l(\beta)}{\partial \beta} = X^T (y - p)$$

$$\blacktriangleright \frac{\partial^2 l(\beta)}{\partial \beta \partial \beta^T} = -X^T W X$$

Newton-Raphson updates

$$\blacktriangleright \beta^{new} = \beta^{old} + (X^T W X)^{-1} X^T (y - p) = (X^T W X)^{-1} X^T W (X \beta^{old} + W^{-1} (y - p)) = (X^T W X)^{-1} X^T W z$$

$$\blacktriangleright z = X \beta^{old} + W^{-1} (y - p), \text{ also called adjusted response}$$

$\blacktriangleright$  Iteratively reweighted least squares

$$\blacktriangleright \text{At each iteration } \beta^{new} \leftarrow \operatorname{argmin}_{\beta} (z - X\beta)^T W (z - X\beta)$$

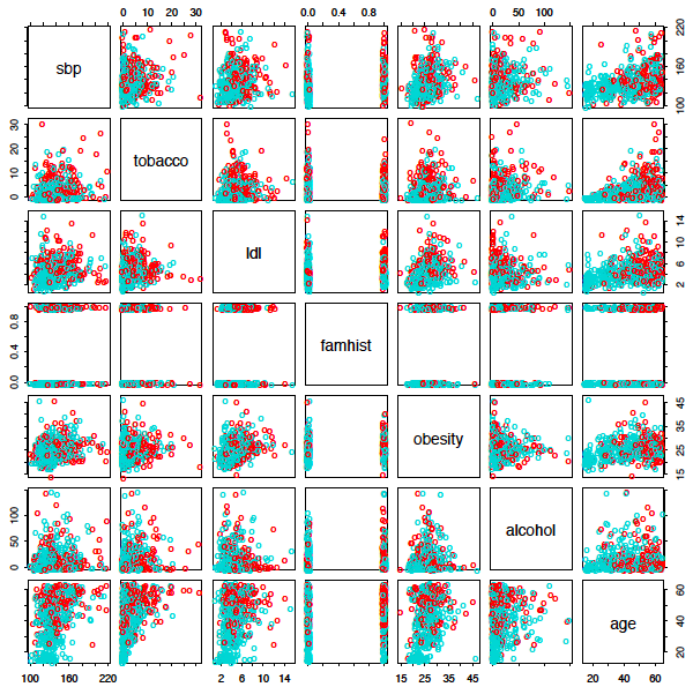
# Example - South Africa Heart Disease

## Aim

- ▶ Establish the intensity of ischemic heart disease risk factors in the high-incidence region (Western Cape, South Africa)

## Data

- ▶ Participants: white males between 15 and 64
- ▶ Output: presence or absence of myocardial infarction (MI) at the time of the survey (binary response)
- ▶ Descriptive statistics: overall prevalence of MI: 5.1% in that region, while 160 cases and 302 controls in this data set



# Example - Logistic Regression Model Fit

|             | Coefficient | Std. Error | Z Score |
|-------------|-------------|------------|---------|
| (Intercept) | -4.130      | 0.964      | -4.285  |
| sbp         | 0.006       | 0.006      | 1.023   |
| tobacco     | 0.080       | 0.026      | 3.034   |
| ldl         | 0.185       | 0.057      | 3.219   |
| famhist     | 0.939       | 0.225      | 4.178   |
| obesity     | -0.035      | 0.029      | -1.187  |
| alcohol     | 0.001       | 0.004      | 0.136   |
| age         | 0.043       | 0.010      | 4.184   |

Any surprises?

- ▶ Which factors are likely to contribute to heart disease, based on common sense?
- ▶ sbp? obesity?
- ▶ Alternatives?

## Example - Variable Selection

|             | Coefficient | Std. Error | Z score |
|-------------|-------------|------------|---------|
| (Intercept) | -4.204      | 0.498      | -8.45   |
| tobacco     | 0.081       | 0.026      | 3.16    |
| ldl         | 0.168       | 0.054      | 3.09    |
| famhist     | 0.924       | 0.223      | 4.14    |
| age         | 0.044       | 0.010      | 4.52    |

Find variables that are sufficient for explaining their joint effect on the prevalence of heart disease.

- ▶ Successively drop variables that are not "significant"
- ▶ How to interpret coefficients?
- ▶ How to calculate confidence intervals?

# L1 Regularized Logistic Regression

# L1 Regularized Logistic Regression

Maximize a penalized version of the log likelihood

$$\max_{\beta_0, \beta} \left\{ \sum_{i=1}^N [y_i(\beta_0 + \beta^T x_i) - \log(1 + e^{\beta_0 + \beta^T x_i})] - \lambda \sum_{j=1}^p |\beta_j| \right\}$$

- ▶ Standardize predictors first, don't penalize intercept
- ▶ Penalized likelihood surface is concave, solutions can be found by nonlinear programming methods
- ▶ Quadratic approximation using Newton-Raphson by repeated application of a weighted lasso algorithm

# Logistic Regression or LDA?

## LDA decision boundary

$$\begin{aligned} \text{▶ } \log \frac{Pr(G=k|X=x)}{Pr(G=K|X=x)} &= \\ \log \frac{\pi_k}{\pi_K} - \frac{1}{2}(\mu_k + \mu_K)^T \Sigma^{-1}(\mu_k - \mu_K) + x^T \Sigma^{-1}(\mu_k - \mu_K) &= \alpha_{k0} + \alpha_k^T x \end{aligned}$$

## Logistic regression

$$\text{▶ } \log \frac{Pr(G=k|X=x)}{Pr(G=K|X=x)} = \beta_{k0} + \beta_k^T x$$

Difference?  $Pr(X, G = k) = Pr(X)Pr(G = k | X)$

- ▶ LDA is fit by maximizing the full likelihood  
 $Pr(X, G = k) = \phi(X; \mu_k, \Sigma)\pi_k$
- ▶ Logistic regression leaves the marginal density of  $X$  as an arbitrary, and is fit by maximizing the conditional likelihood  $Pr(G = k | X)$

# Logistic Regression or LDA?

The marginal density in LDA:  $Pr(X) = \sum_{k=1}^K \pi_k \phi(X; \mu_k, \Sigma)$

If  $f_k(x) \sim \mathcal{N}$

- ▶ LDA gives more info about the parameters
- ▶ LDA can estimate parameters more efficiently (lower variance).

Bad things about LDA...

- ▶ Less robust to gross outliers
- ▶ Observations far from decision boundary influence  $\hat{\Sigma}$
- ▶ observations without class labels have information about the parameters
- ▶ Quantitative variables in  $X$

# Naive Bayes

From Bayes Theorem

$$P(Y = k \mid x_1, \dots, x_p) \propto \pi_k P(x_1, \dots, x_p \mid Y = k)$$

Naive Bayes assumption: conditional independence of  $x_j$ 's given  $y$

$$P(x_1, \dots, x_p \mid Y = k) = \prod_{j=1}^p P(x_j \mid Y = k)$$

The relationship is thus simplified to

$$P(y \mid x_1, \dots, x_p) \propto \pi_k \prod_{j=1}^p P(x_j \mid y)$$

# Naive Bayes

Therefore, the classification rule is

$$\hat{y} = \operatorname{argmax}_k \pi_k \prod_{j=1}^p P(x_j \mid Y = k)$$

- ▶ Despite apparently over-simplified assumptions, naive Bayes classifiers have worked quite well in many real-world situations (document classification and spam filtering)
- ▶ Note: Naive Bayes is known as a decent classifier, but a bad estimator (probability outputs are not to be taken too seriously).

# Gaussian Naive Bayes

The likelihood of the feature is assumed to be Gaussian

$$P(X_j \mid Y = k) = \frac{1}{\sqrt{2\pi\sigma_k^2} \exp\left(-\frac{(x_j - \mu_k)^2}{2\pi\sigma_k^2}\right)}$$

Parameters  $\sigma_k$  and  $\mu_k$  are estimated using maximum likelihood method.

# Multinomial Naive Bayes

Often used in text classification

- ▶ Data are typically represented as word vector counts

Parameter vector  $\theta_k = (\theta_{k1}, \dots, \theta_{kp})$  for each class  $k$

- ▶  $p$ : the number of features (the size of vocabulary)
- ▶  $\theta_{kj}$ :  $P(x_j \mid Y = k)$ , the probability of feature  $j$  appearing in a sample belonging to class  $k$

$\theta_k$  can be estimated by smooth version of maximum likelihood

$$\hat{\theta}_{kj} = \frac{N_{kj} + \alpha}{N_k + \alpha p}$$

# Multinomial Naive Bayes

$$\hat{\theta}_{kj} = \frac{N_{kj} + \alpha}{N_k + \alpha p}$$

- ▶  $N_{kj}$ : number of times feature  $j$  appears in class  $k$  in the training set
- ▶  $N_k$ : total count of all features for class  $k$  in the training set
- ▶  $\alpha > 0$ : smoothing priors, prevent zero probabilities for feature not present in the training set
- ▶  $\alpha = 1$  : Laplace smoothing
- ▶  $\alpha < 1$  : Lidstone smoothing

# Complement Naive Bayes

- ▶ Adaptation of the standard Multinomial Naive Bayes algorithm
- ▶ Suited for **imbalanced data sets**
- ▶ Uses statistics from the complement of each class to compute model's weights
- ▶ Parameter estimates of CNB are more stable than those for MNB
- ▶ CNB regularly outperforms MNB on text classification

# Complement Naive Bayes

$$\hat{\theta}_{kj} = \frac{\alpha_j + \sum_{i:y_i \neq k} d_{ij}}{\alpha + \sum_{i:y_i \neq k} \sum_l d_{il}}$$

- ▶  $d_{ij}$ : count of term  $j$  in document  $i$
- ▶  $\alpha = \sum_j \alpha_j$
- ▶ summation over all documents  $i$  not in class  $k$

$$w_{kj} = \frac{\log \hat{\theta}_{kj}}{\sum_l \log \hat{\theta}_{kl}}$$

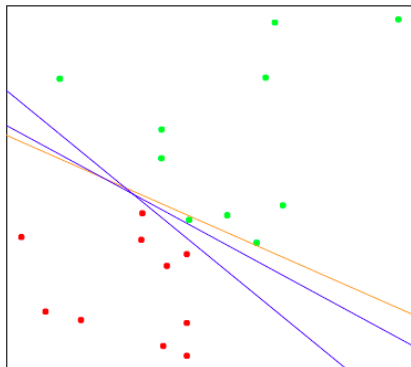
$$\hat{k} = \operatorname{argmin}_k \sum_j t_j w_{kj}$$

i.e., Classification rule: assign document to the class that is the poorest complement match

# Separating Hyperplanes

# Separating Hyperplanes

20 data points  
two classes in  $\mathbb{R}^2$ .



## Blue lines

- ▶ infinitely many possible separating hyperplanes

## Orange line

- ▶ linear decision boundary given by least squares

## Perceptrons

- ▶ compute a linear combination of the input features and return the sign
- ▶  $\{x : \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 = 0\}$

# Rosenblatt's Perceptron Learning Algorithm

Find a separating hyperplane by minimizing the distance of misclassified points to the decision boundary

$$D(\beta, \beta_0) = -\sum_{i \in \mathcal{M}} y_i (\beta_0 + X_i^\top \beta)$$

- ▶ If  $y_i = 1$  is misclassified,  $\beta_0 + X_i^\top \beta < 0$
- ▶ If  $y_i = -1$  is misclassified,  $\beta_0 + X_i^\top \beta > 0$
- ▶  $\mathcal{M}$  indexes the set of misclassified points (assumed fixed)

The quantity  $D(\beta, \beta_0)$  is non-negative and proportional to the distance of the misclassified points to the decision boundary defined by  $\beta_0 + X_i^\top \beta = 0$

# Parameter Estimation for Perceptron

Gradient is given by

$$\blacktriangleright \frac{\partial D(\beta, \beta_0)}{\partial \beta} = - \sum_{i \in \mathcal{M}} y_i x_i$$

$$\blacktriangleright \frac{\partial D(\beta, \beta_0)}{\partial \beta_0} = - \sum_{i \in \mathcal{M}} y_i$$

Stochastic gradient descent

$$\begin{bmatrix} \beta \\ \beta_0 \end{bmatrix} \leftarrow \begin{bmatrix} \beta \\ \beta_0 \end{bmatrix} + \rho \begin{bmatrix} y_i x_i \\ y_i \end{bmatrix}$$

- ▶ A step is taken after each observation is visited
- ▶ Along the negative gradient direction
- ▶ Repeat until no points are misclassified
- ▶ Step size  $\rho$
- ▶ More efficient than regular Gradient Descent

# Problems...

Details summarized in Ripley (1996)

- ▶ All separating hyperplanes are considered equally valid
- ▶ When the data are separable, there are many solutions, and which one is found depends on the starting values
- ▶ The “finite” number of steps can be very large. The smaller the gap, the longer the time to find it
- ▶ When the data are not separable, the algorithm will not converge, and cycles develop. The cycles can be long and therefore hard to detect